

	1	2	3	4	5	6	7	8
A	AUDIO EXPANSION CONN File: AUDIO_EXPANSION_CONN.kicad_sch	SERIAL EXPANSION CONN File: SERIAL_EXPANSION_CONN.kicad_sch	I2S HEADER File: I2S_HEADER.kicad_sch	AUDIO DAC PCM1690 File: AUDIO_DAC_PCM1690.kicad_sch	DAC0 LPF & BUFFERS File: DAC0_LPF_and_BUFFERS.kicad_sch	DAC1 LPF & BUFFERS File: DAC1_LPF_and_BUFFERS.kicad_sch	AUDIO ADC PCM1865 File: AUDIO_ADC_PCM1865.kicad_sch	
B	DIGITAL AUDIO RECEIVER File: DIGITAL_AUDIO_RECEIVER.kicad_sch	DIR OPTICAL & COAXIAL INPUT File: DIR_OPTICAL_and_COAXIAL_INPUT.kicad_sch	OPAMP POWER File: OPAMP_POWER.kicad_sch					
C								
D								
E								
F								

NDH

Sheet: /
File: Audio_Carte_fille.kicad_sch

Title: Carte fille (Base Saine)

Size: A3Date: 2025-06-09KiCad E.D.A. 9.0.0

Rev: 1.0Id: 1/11

	5	6	7	8
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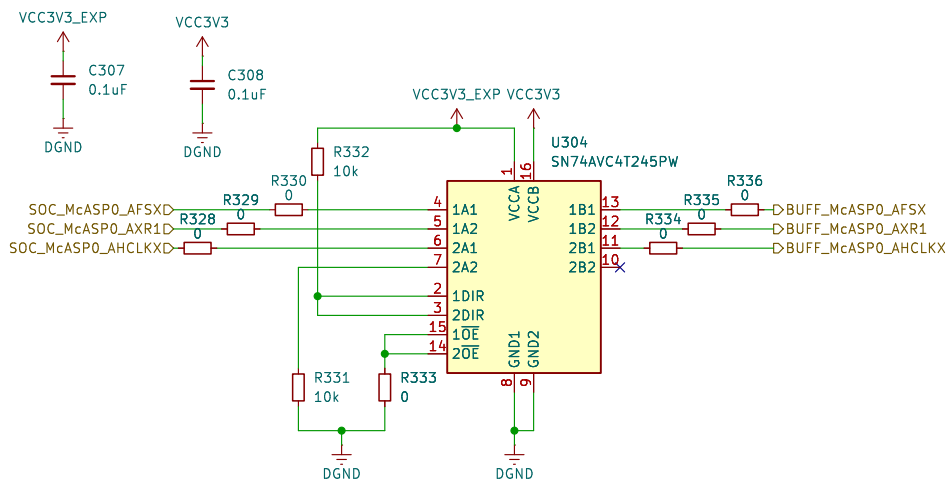
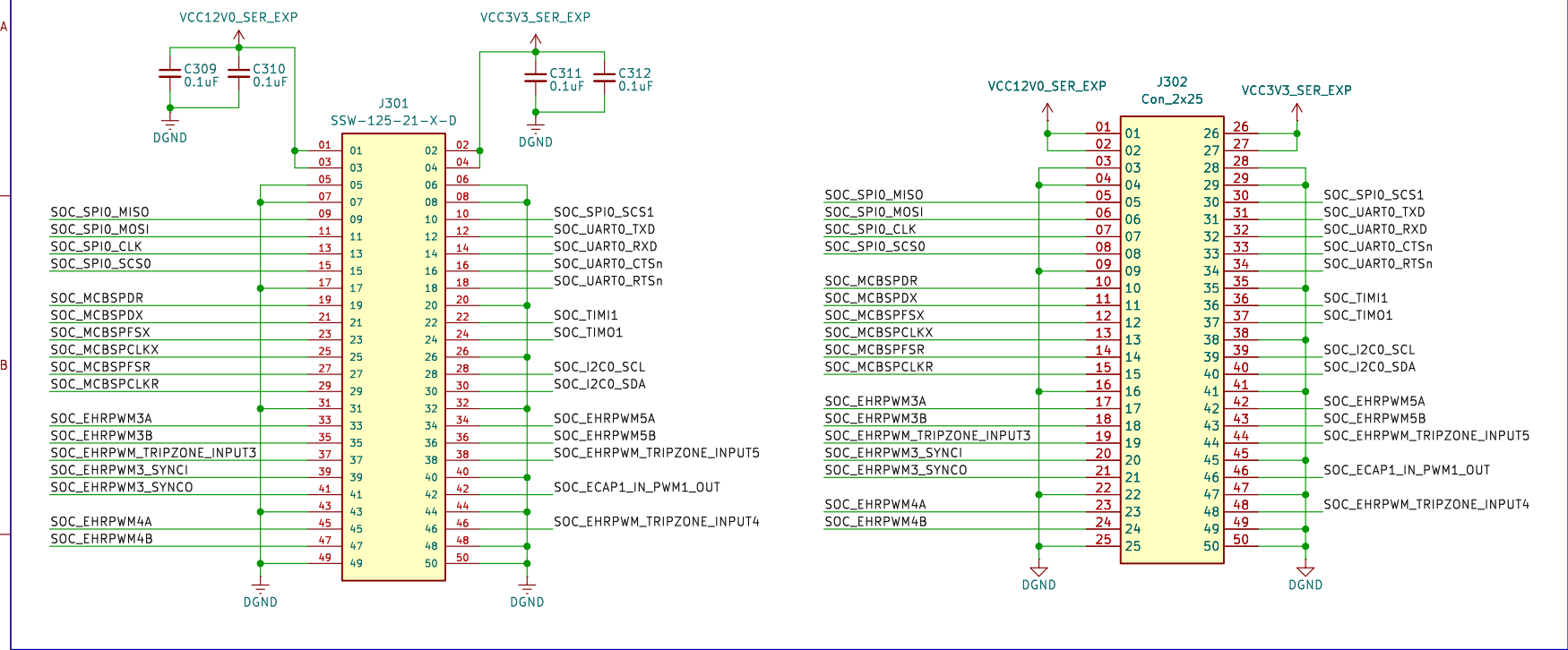
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Title: Carte fille (Base Saine)

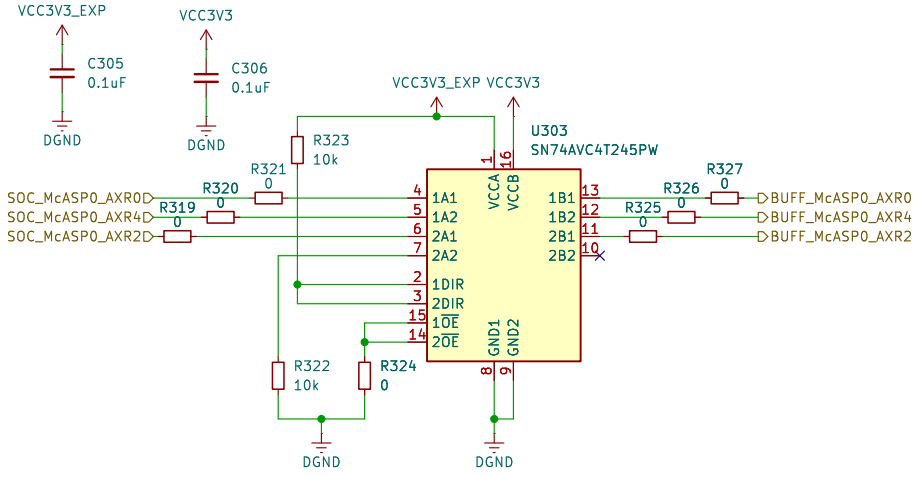
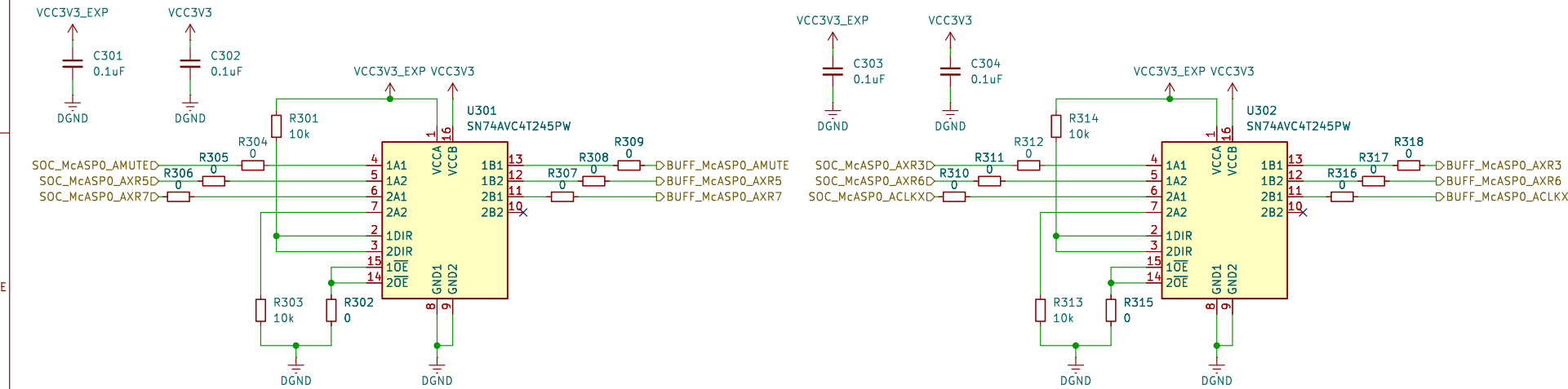
SIZE: AS	DATE: 2025-05-05	REV: 1.0
KiCad E.D.A. 9.0.0		Id: 2/11

	7	8
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SERIAL EXPANSION CONN



DAC CLOCK AND DATA BUFFER



NDH

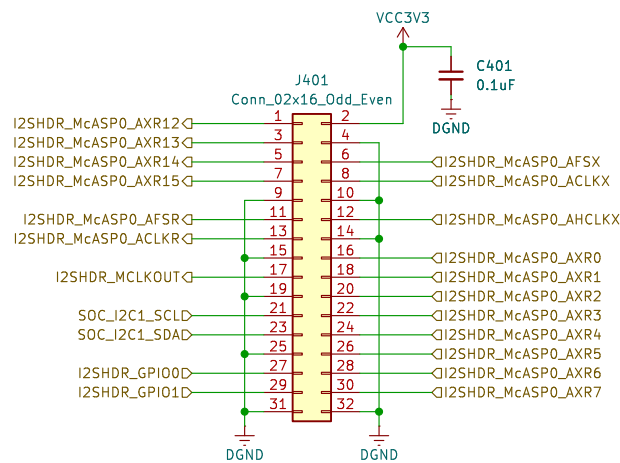
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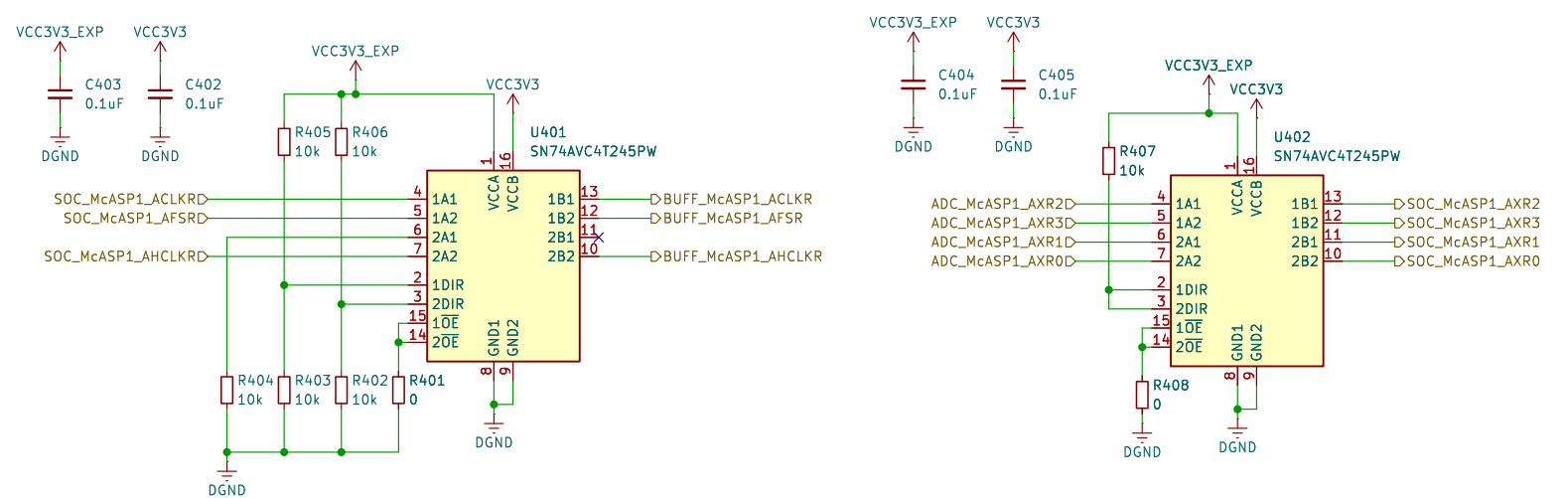
Size: A3 Date: 2025-06-09
KiCad E.D.A. 9.0.0

Rev: 1.0
Id: 3/11

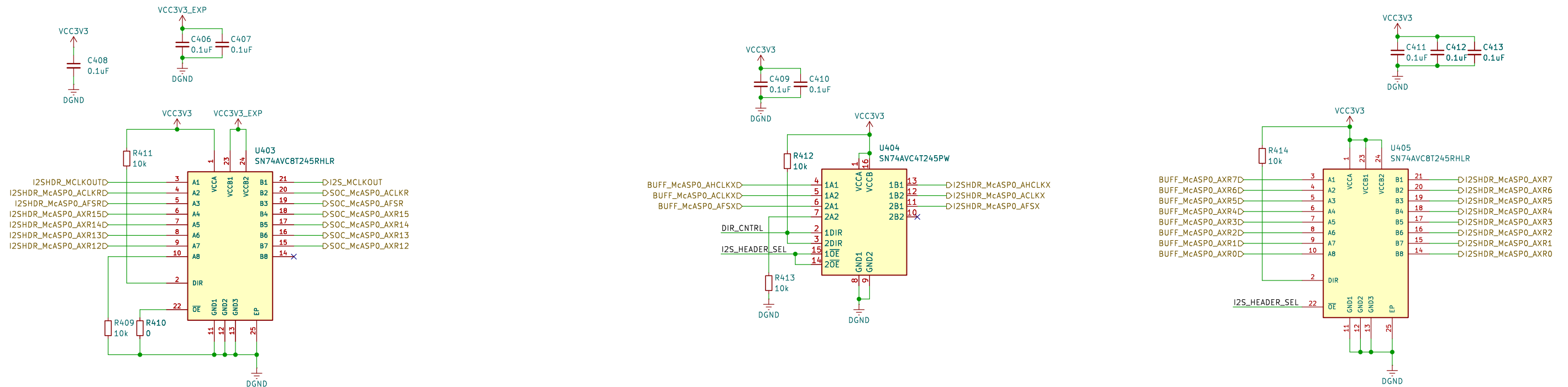
I2S HEADER



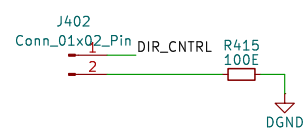
ADC CLOCK & DATA BUFFER



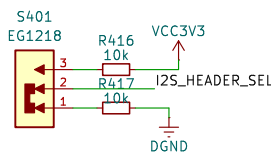
I2S CLOCK AND DATA BUFFER



INSTALL JUMPER FOR TX SLAVE MODE



I2S HEADER SELECTION SWITCH



NDH

Sheet: /I2S HEADER/
File: I2S_HEADER.kicad_sch

Title: Carte fille (Base Saine)

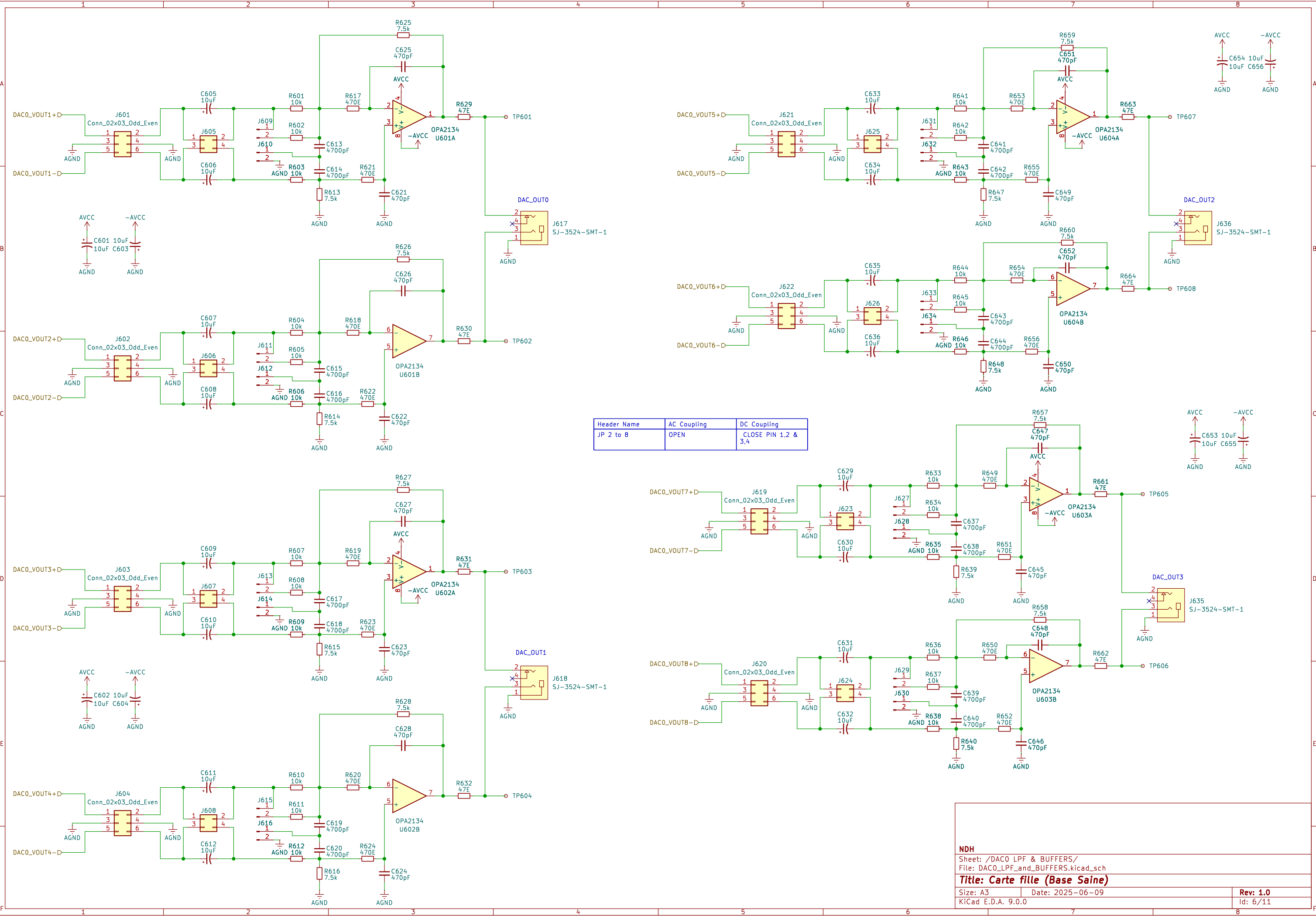
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Date: 2025-06-09

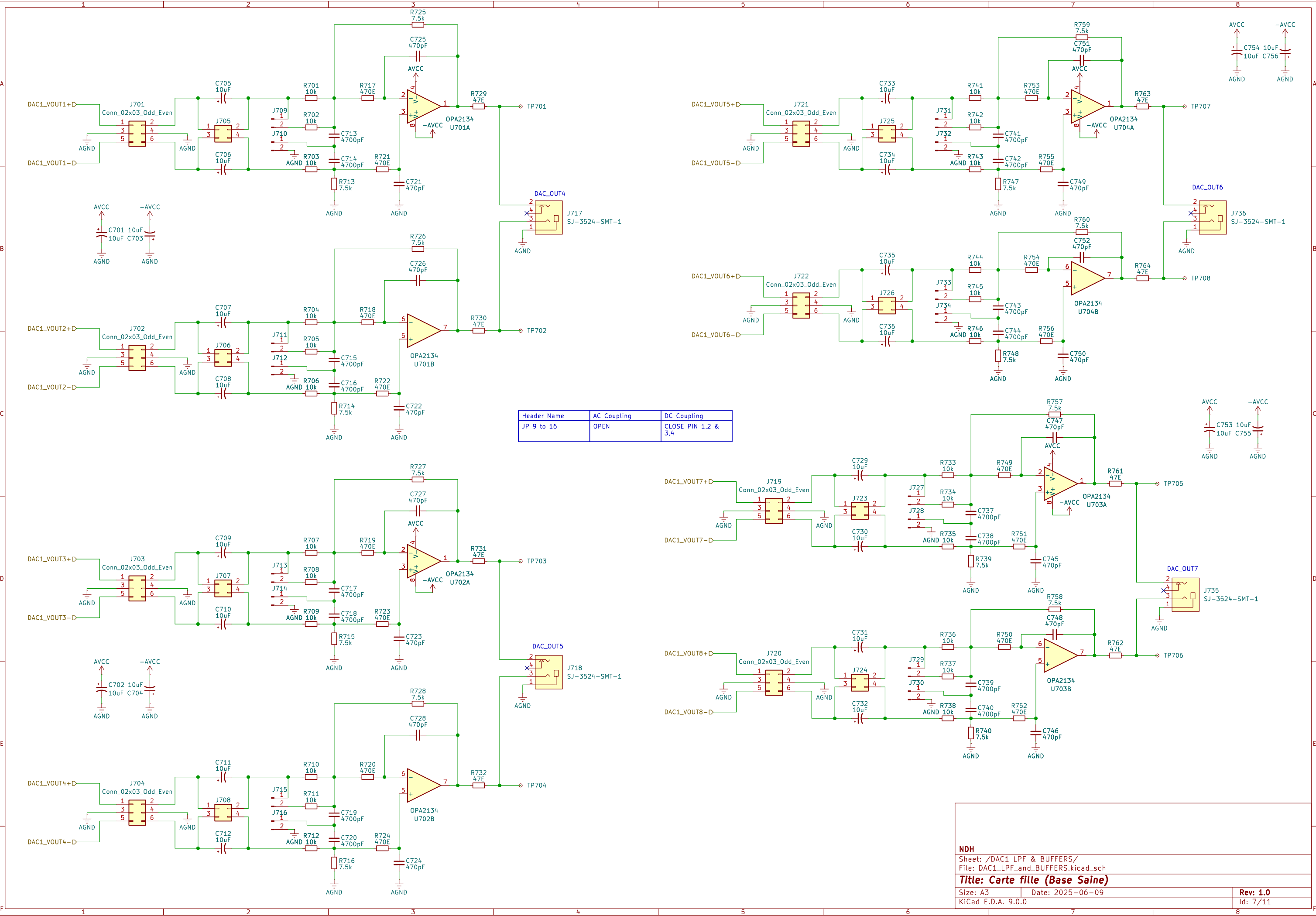
Rev: 1.0

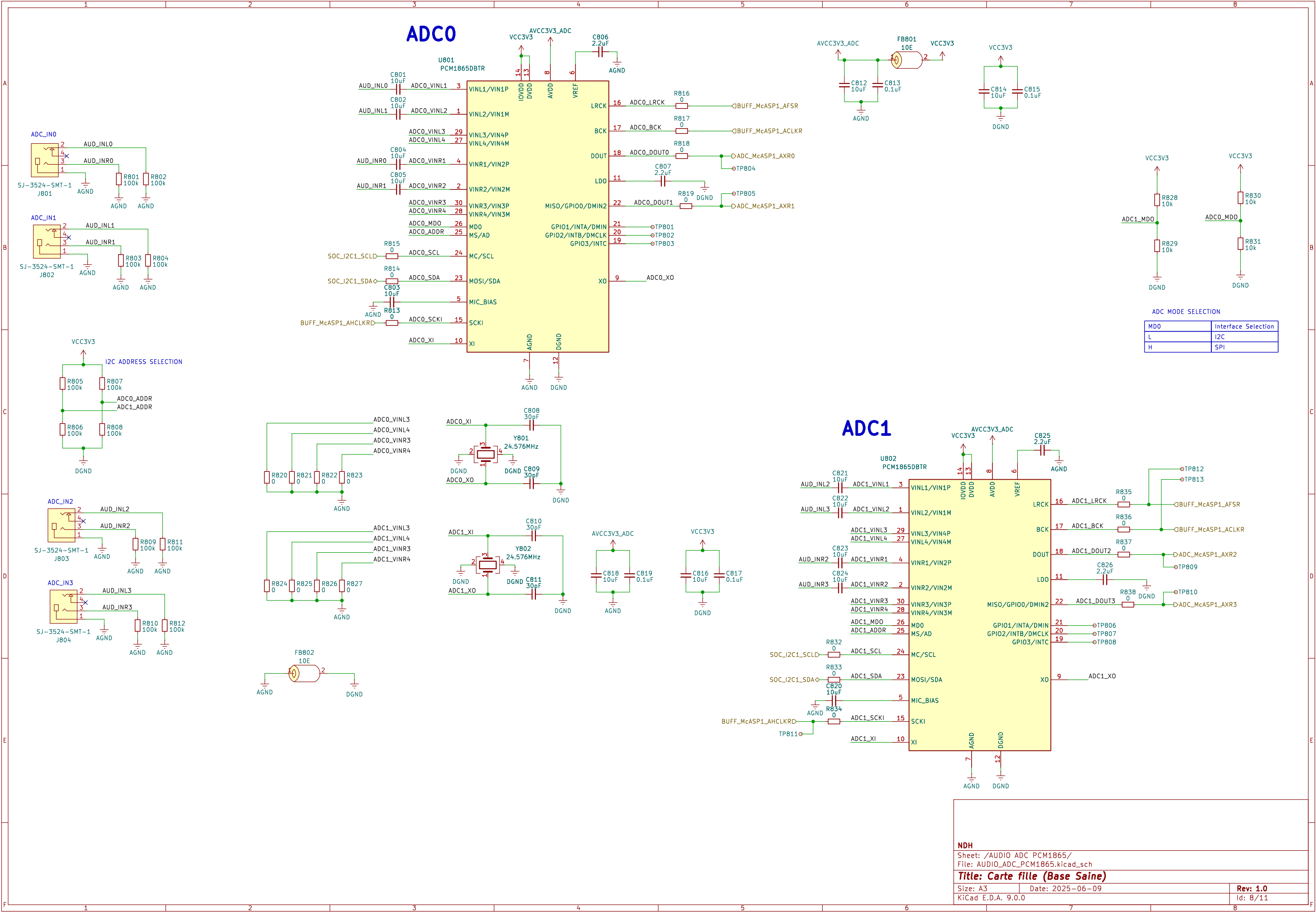
KiCad E.D.A. 9.0.0

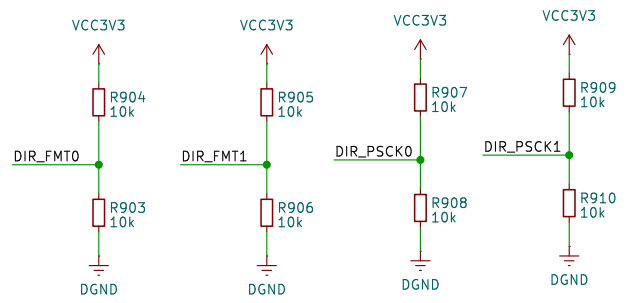
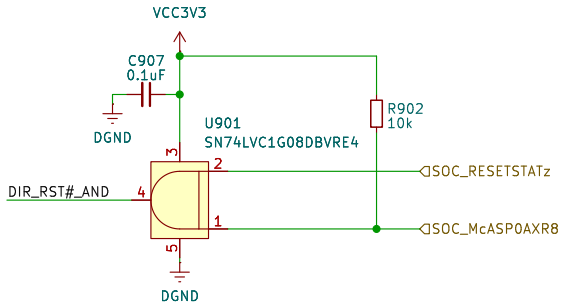
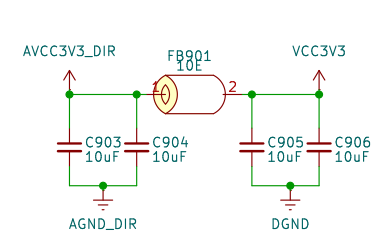
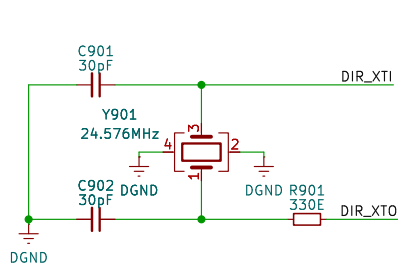
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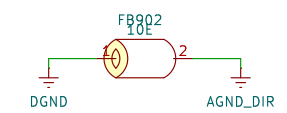
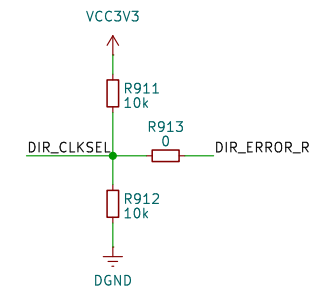
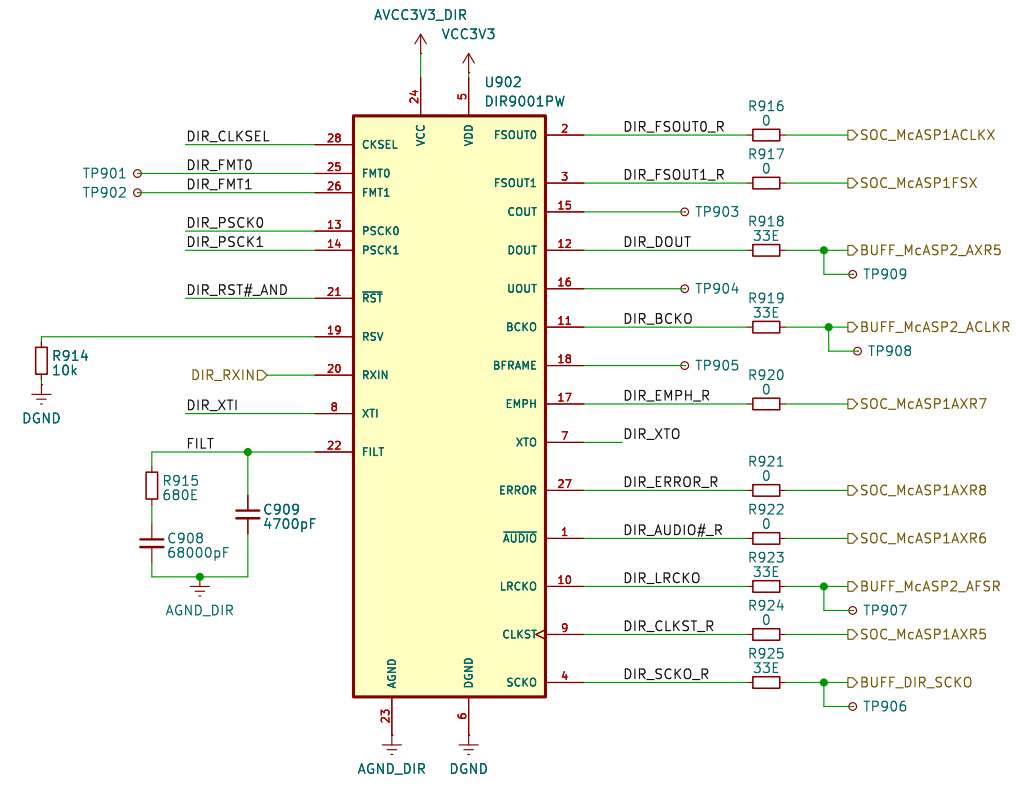
Header Name	AC Coupling	DC Coupling
JP 2 to 8	OPEN	CLOSE PIN 1,2 & 3,4







DIR9001



DIR Operation Mode and Clock Source

Operation Mode	CLKSEL	ERROR Pin Status	SCKO, BCKO, LRCKO CLOCK SOURCE
PLL	L	H	PLL (VCO) free-running Clock
		L	PLL recovered clock
XTAL	H	H	XTAL clock
		L	XTAL clock
AUTO	Connected to ERRO	H	XTAL clock
		L	PLL recovered clock

SYSTEM CLK FREQUENCY OF PLL SELECTION

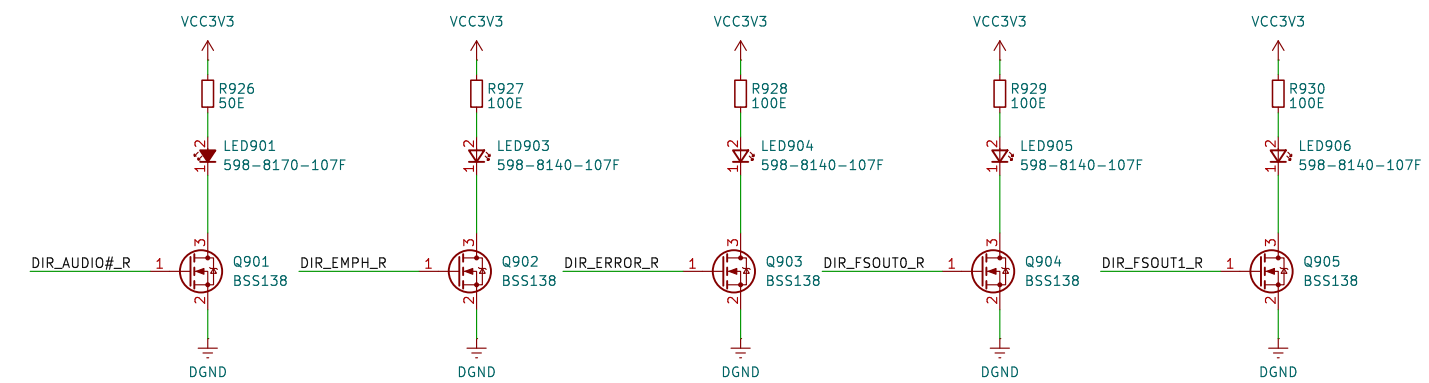
PSCK[1:0] SETTING		OUTPUT CLOCK FROM PLL SOURCE		
PSCK1	PSCK0	SCKO	BCKO	LRCKO
L	L	128 fs	64 fs	fs
L	H	256 fs	64 fs	fs
H	L	384 fs	64 fs	fs
H	H	512 fs	64 fs	fs

SERIAL AUDIO DATA OUTPUT FORMAT SELECTION

FMT[1:0] SETTINGS		AL AUDIO DATA OUTPUT FORMAT
FMT1	FMT0	
L	L	16-bit, MSB-first, right-justified
L	H	24-bit, MSB-first, right-justified
H	L	24-bit, MSB-first, left-justified
H	H	24-bit, MSB-first, I2S

LED STATUS TABLE

STATUS	LED1	LED2	LED3	LED4	LED5
Non Audio Sample Word	ON	OFF	OFF	OFF	OFF
Pre-Emphasis (50 ms/15 ms)	OFF	ON	OFF	OFF	OFF
Internal PLL or Data parity error	OFF	OFF	ON	OFF	OFF
Sample Frequency Result Output 0	OFF	OFF	OFF	ON	OFF
Sample Frequency Result Output 1	OFF	OFF	OFF	OFF	ON



NDH

Sheet: /DIGITAL AUDIO RECEIVER/
File: DIGITAL_AUDIO_RECEIVER.kicad_sch

Title: Carte fille (Base Saine)

Size: A3

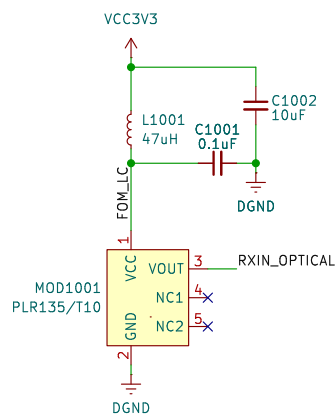
Date: 2025-06-09

Rev: 1.0

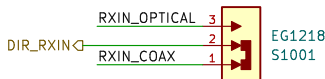
KiCad E.D.A. 9.0.0

Id: 9/11

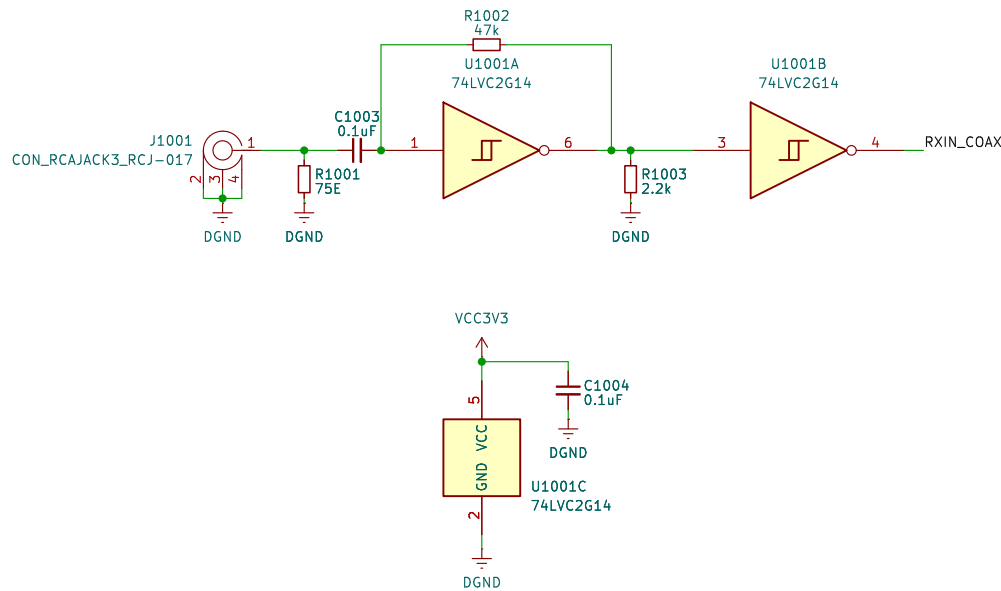
FIBER OPTIC RECEIVER MODULE



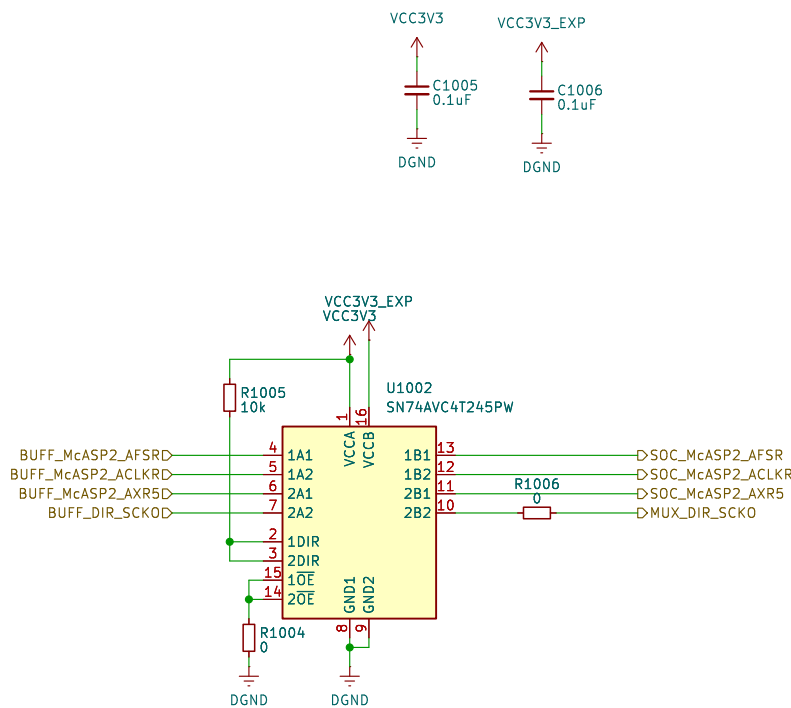
INPUT SELECTION SWITCH



COAXIAL RCA CONNECTOR & LINE DRIVER



DIR CLOCK AND DATA BUFFER



VCC12V_EXP

C1109
4.7uF

DGND

SRF0905-471Y
L1103

MOD1101
JCY1012D15

+VIN
-VIN

+VOUT
COMMON1
COMMON2
-VOUT

AVCC
-AVCC
AGND

TP1104
TP1105

TP1106
TP1107
TP1108
TP1109
DGND

Id: 11/11